Lecture 15

Tuesday, 24 October 2023 10:40 AM

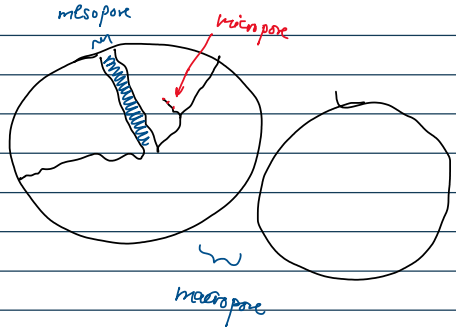
Characterisation Methods for Surface area & pore size determination

Micropore, Mesopore, Macropore

Adsorption isotherm

Small angle X-ray scattering:

Mercury porosimetry: non-wetting liquid → pressure needed to fill the pore space



Gas adsorption: surface atoms → unbalanced forces / undercoordinated bonds

Physisorption forces: + Van der waal's forces → dispersion force
→ ion-dipole
→ ion-induced dipole
→ Dipole-dipole

→ non-specific (Can adsorb anywhere on the surface)!

→ How do you determine if it is blocking multiple sites??

→ Can also have multilayer (partial coverages)!

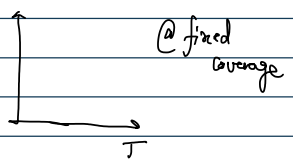
→ But may not be layer by layer } Need a mathematical model

All equilibrium measurements

→ isobar (p = const)

→ isotherm (T = constant) → coverage as function of increasing pressure

→ isostere (Coverage constant)



Use different adsorbing molecules and then coverage will be off by factor of molecule volume